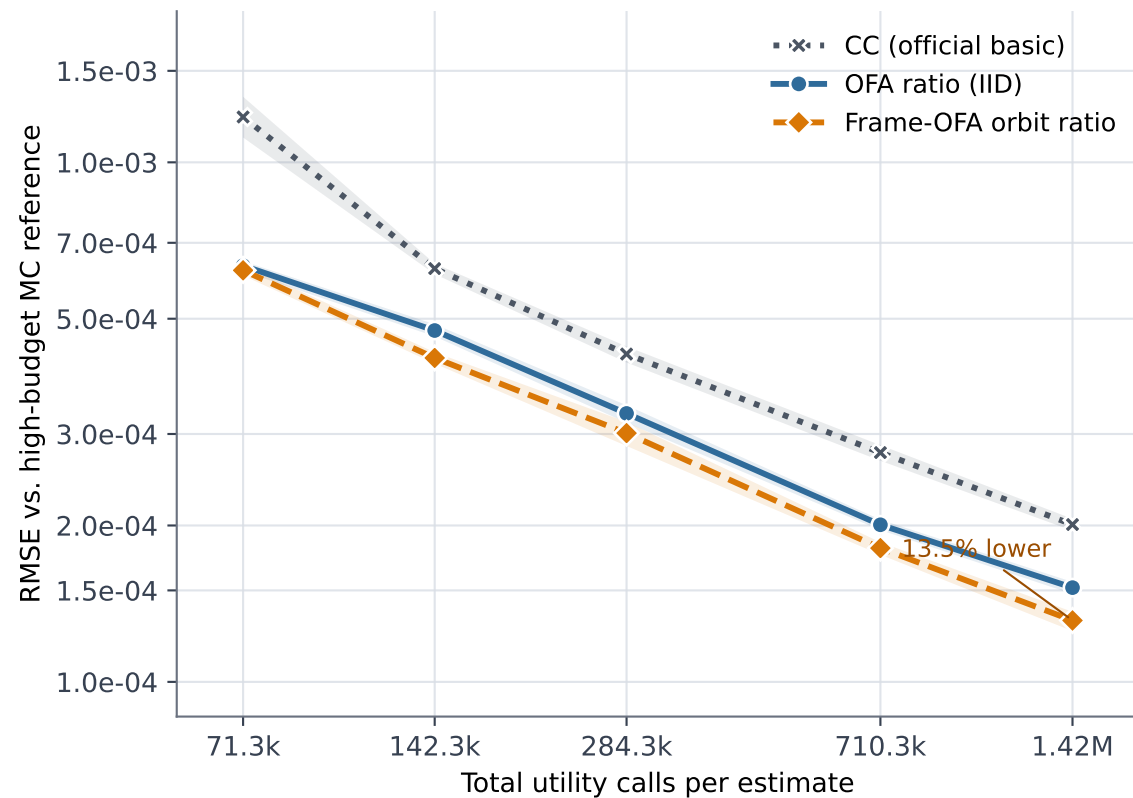


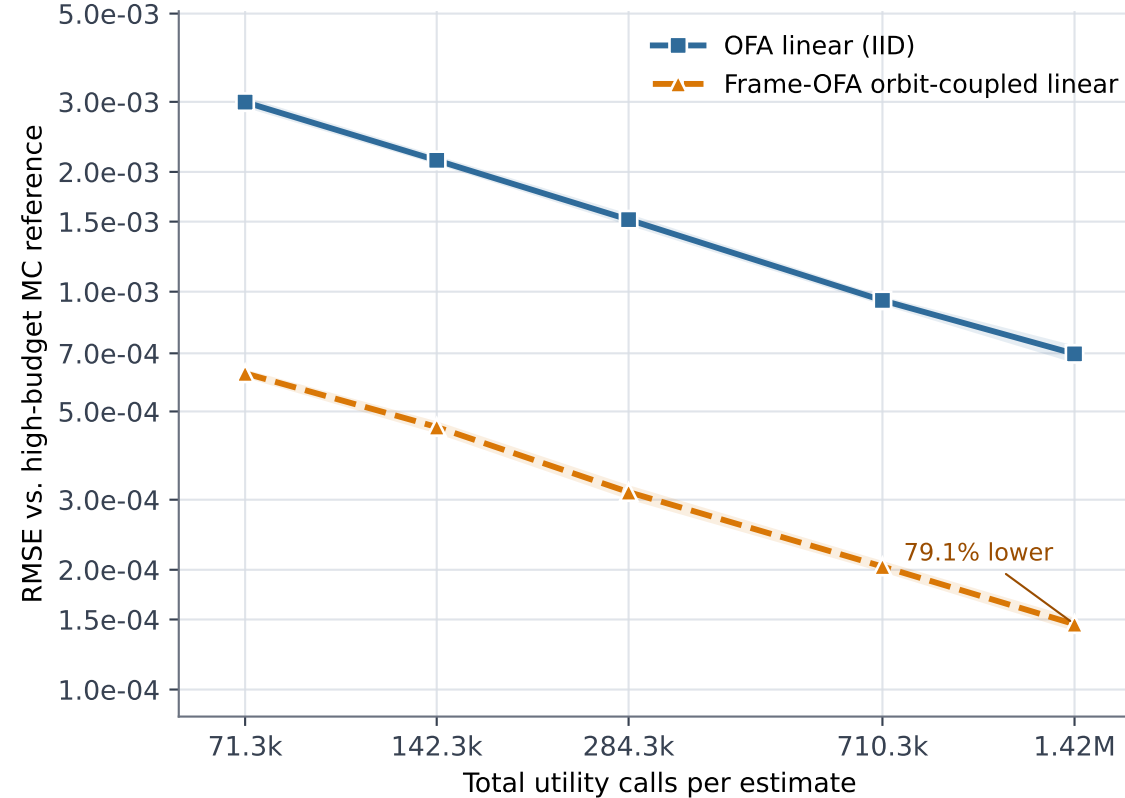
Wine data Shapley RMSE vs. total utility calls

142 training points as players · 36 fixed test points · RBF-SVM accuracy · 10 repeats per budget · 71.3k–1.42M total calls

Stratified-mean / ratio estimators



Linear estimators



Shaded bands: 95% repeat bootstrap intervals. Both axes use logarithmic scales. OFA totals include 286 exact boundary calls; CC uses each total entirely as two-call complement pairs. Ground-truth SE-RMSE: 3.33e-05.